HARPER KNITTEL

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RESEARCH INTERESTS

My research focuses on graph algorithms for fairness and scalability on massive graphs. Some of my previous and ongoing works include: (1) adaptive and non-adaptive massively parallel matching, edge coloring, clustering, minimum cut, and geometric embedding, (2) fair hierarchical clustering and resource allocation, and (3) incentive structures in multi-agent games of influence and matching markets.

ACADEMIC POSITIONS

Assistant Professor — 2025 - Present
Reed College — *Portland, OR*

Postdoctoral Researcher — 2024
University of California, San Diego — *La Jolla, CA*
Advisors: Barna Saha and Sanjoy Dasgupta

EDUCATION

University of Maryland, College Park — College Park, MD
PhD in Computer Science, 3.97 GPA — *Dec 2023*
MS in Computer Science — *May 2020*
Dissertation: Fair and Scalable Algorithms on Massive Graphs
Defense Committee: MohammadTaghi Hajiaghayi (chair, advisor), John Dickerson (advisor), Ravi Kumar, David Mount, Alexander Barg

Harvey Mudd College — Claremont, CA
B.S. in Computer Science and Mathematics, 3.75 GPA — *May 2018*
High Distinction, Honors in Math and Computer Science, Dean's List

HONORS AND AWARDS

Larry S. Davis Dissertation Award 2023
Jane Street Graduate Research Fellowship Finalist 2023
NIH Summer Cancer Research Training Award 2023
Meta Research PhD Fellowship Finalist 2022
ARCS Endowment Award .... 2021 - 2022
Ann G Wylie Dissertation Fellowship 2021
Dean's Fellow .... 2018 - 2020
HMC Class of '94 Award 2018
Sandra Forsythe Memorial Scholarship 2014

TEACHING

HMC = Harvey Mudd College, UMD = University of Maryland

Reed College	Algorithms & Data Structures	CSCI/MATH 382	F26, S26, F25
	Computer Science II	CSCI 221	F26, F25, S25
	Advanced Algorithms	CSCI/MATH 382	S26
	Mechanism Design	CSCI 375	S25
UMD	Mechanism Design	CMSC 498T	Sp22

INTERNSHIPS AND RESEARCH POSITIONS

** Denotes remote position.*

Research Fellow, The National Institute of Health, Bethesda MD ... May - Aug 23
Advisor: Cenk Sahinalp

Research Intern, Google, Mountain View CA ... May - Aug 22

Research Intern, Toyota Technological Institute at Chicago, Chicago IL* ... Jun - Aug 21
Advisors: Avrim Blum and Saeed Seddighin

Research Intern, Amazon, Seattle WA* ... Jun - Aug 20

Software Engineering Intern, Google, Seattle WA ... May - Aug 19

Software Engineering Intern, Meta, Menlo Park CA ... May - Jul 18

Project Manager and Researcher, NASA Ames and HMC, Claremont CA ... Aug 17 - May 18
Advisors: James Boerkoel and Jeremy Frank

REU Scholar, Rutgers University, Piscataway NJ ... May - Aug 17
Advisor: Eric Allender

Software Engineering Intern, Bloomberg, New York NY ... Jun - Aug 16

Student Researcher, HMC, Claremont CA ... May 15 - May 18
Advisors: Jessica Wu and Colleen Lewis

Web Development Intern, Napses, Santa Barbara CA* ... May - Aug 14

PUBLICATIONS AND PRESENTATIONS

*[abc]Denotes that authors are listed **alphabetically**. This is the convention in theoretical computer science. Computer science traditionally uses competitive conferences (15-30% accepted) as the main publication venue.*

Note: Publications are listed under my legal name, Marina Knittel.

Conference Publications:

1. Harper Knittel, Max Springer, John Dickerson, and MohammadTaghi Hajiaghayi. “Fair, Polylog-Approximate Low-Cost Hierarchical Clustering”. The Conference on Neural Information Processing Systems (NeurIPS), 2023.

2. Harper Knittel, Max Springer, John Dickerson, and MohammadTaghi Hajiaghayi. “Generalized Reductions: Making any Hierarchical Clustering Fair and Balanced with Low Cost”. International Conference on Machine Learning (ICML), 2023.

3. [abc]AmirMohsen Ahanchi, Alexandr Andoni, MohammadTaghi Hajiaghayi, Harper Knittel, and Peilin Zhong, “Massively Parallel Tree Embeddings for High Dimensional Spaces”. Symposium on Parallelism and Architectures (SPAA), 2023.

4. [abc]MohammadTaghi Hajiaghayi, Harper Knittel, Jan Olkowski, and Hamed Saleh, “Adaptive Massively Parallel Algorithms for Cut Problems”. Symposium on Parallelism and Architectures (SPAA), 2022.

5. Harper Knittel, Samuel Dooley, and John P. Dickerson, “The Dichotomous Affiliate Stable Matching Problem: Approval-Based Matching with Applicant-Employer Relations”. International Joint Conference on Artificial Intelligence (IJCAI), 2022.

6. [abc]MohammadTaghi Hajiaghayi, Harper Knittel, Hamed Saleh, and Hsin-Hao Su, “Adaptive Massively Parallel Constant-round Tree Contraction”. Innovations in Theoretical Computer Science (ITCS), 2022.

7. [abc]Fotini Christia, Michael Curry, Constantinos Daskalakis, Erik Demaine, John P. Dickerson, MohammadTaghi Hajiaghayi, Adam Hesterberg, Harper Knittel, and Aidan Millif, “Scalable Equilibrium Computation in Multi-agent Influence Games on Networks”. The Association for the Advancement of Artificial Intelligence (AAAI), 2021.

8. [abc]Sara Ahmadian, Alessandro Epasto, Harper Knittel, Ravi Kumar, Mohammad Mahdian, Benjamin Moseley, Sergei Vassilvitskii, Philip Pham, and Yuyan Wang. “Fair Hierarchical Clustering”. The Conference on Neural Information Processing Systems (NeurIPS), 2020.

9. [abc]MohammadTaghi Hajiaghayi and Harper Knittel, “Matching Affinity Clustering: Improved Hierarchical Clustering at Scale with Guarantees”. The International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2020. Extended abstract.

10. [abc]Soheil Behnezad, Mahsa Derakhshan, MohammadTaghi Hajiaghayi, Harper Knittel, and Hamed Saleh, “Streaming and Massively Parallel Algorithms for Edge Coloring”. The 27th Annual European Symposium on Algorithms (ESA), 2019.

11. Jordan R. Abrahams, David A. Chu, Grace Diehl, Harper Knittel, Judy Lin, William Lloyd, James C. Boerkoel Jr., and Jeremy Frank, “DREAM: An Algorithm for Mitigating the Overhead of Robust Rescheduling”. The 29th International Conference on Automated Planning and Scheduling (ICAPS), 2019.

12. Hoaxing Du, Yi Sheng Ong, Harper Knittel, Ross Mawhorter, Ivy Liu, Gianluca Gross, Reiko Tojo, Ran Libeskind-Hadas, and Yi-Chieh Wu, “Multiple Optimal Reconciliations with Gene Duplication, Loss, and Coalescence”. The 17th Asia Pacific Bioinformatics Conference (APBC), 2019.

Invited Presentations:

1. “Fair Hierarchical Clustering.” EnCORE, Theory Seminar. 2023.

2. “Fair Hierarchical Clustering.” Toyota Technological Institute at Chicago, NSF-TRIPODS Postdoc Workshop. 2023.

3. “Adaptive Massively Parallel Constant-round Tree Contraction.” John’s Hopkins University, Theory Seminar. 2021.

4. “Fair Hierarchical Clustering.” The Sets & Partitions Workshop at NeurIPS. 2019

5. “A Cost Function for Hierarchical Clustering.” Google internal seminar. 2019.

SERVICE, WORKSHOPS, AND LEADERSHIP

Program Committee	AAAI, GAIW, NeurIPS	2025
	ICML, IJCAI	2024
	EAAMO	2023
	GAIW	2021-2022
Academic Reviewer	SODA, AAAI, NeurIPS, ESA, ICALP, SPAA, JAIR, SIDMA, Algorithmica	
Other	AAAI Fair Clustering Tutorial Organizer	2022
	Women in Theory Workshop	2022
	Google CSRMP Class of 2021	2021
	CRA-WP Grad Cohort for Women	2021
University of Maryland	Grad CS LGBTQ+ Initiative Founder	2019 - Present
	Executive Committee Member	2018 - Present
	Peer Mentor	2021 - Present
	Capital Area Theory Seminar Organizer	2019 - 2020
	CS Women Mentor	2018 - 2020
	Graduate Admissions Volunteer	2018 - 2019
Harvey Mudd College	Committee for Activities Planning Member	2017 - 2018
	Women in Math Club President	2017 - 2018
	LGBTQ+ Club Mentor	2017 - 2018
	Orientation Leader	2015 - 2017
	Dorm President	2016 - 2017
	Dorm Treasurer	2015 - 2016